

A university gallery app stores photos from seminars, lab sessions, sports events, cultural programmes, and industrial visits. Some photos contain clear faces, while others contain only banners, stage background, lab instruments, playground scenes, or group activities. The system must automatically organize photos into meaningful albums without depending on faces alone. It uses image content, event context, date, location, and visual similarity. Apply the concept and select the most suitable task.

- A. Event-wise grouping
- B. Lens-based correction
- C. Lane-wise detection
- D. Pixel-wise sampling

Correct Answer: A

Explanation:

Photo album organization groups images using visual content, faces, events, time, location, and similarity.

A smart attendance system captures a classroom image during lecture hours. The image contains many students sitting at different distances from the camera. Before identifying the students, the system must first locate every visible face region in the image. These face regions are then cropped and passed to the next module. Apply the concept and select the expected output.

- A. Identity labels
- B. Face boxes
- C. Scene classes
- D. Depth maps

Correct Answer: B

Explanation:

Face detection locates face regions in an image and commonly gives bounding boxes around detected faces.

A research laboratory uses a face-based entry system. The camera has already detected and cropped the face of the person standing near the door. The system now compares the extracted facial features with stored authorized user profiles. Based on the match, it allows or denies entry. Apply the concept and select the main task.

- A. Face locating
- B. Scene grouping
- C. Identity checking
- D. Object tracking

Correct Answer: C

Explanation:

Face recognition verifies or identifies a person by comparing facial features with stored face templates.

A smart gallery creates separate albums for different faculty members and students. First, it scans each image to find face regions. Next, it extracts facial features and compares them across photos. Finally, it places visually matching faces into the same person-wise album. Apply the workflow and select the correct order.

- A. Matching → detection → grouping
- B. Grouping → cropping → matching
- C. Calibration → detection → grouping
- D. Detection → matching → grouping

Correct Answer: D

Explanation:

The correct workflow is face detection, feature matching or recognition, and then photo grouping.

A staff member wants all photos of the same colleague to appear in one album. The photos were taken during different programmes, with changes in dress, background, camera angle, and lighting. Event banners and background objects are not reliable because they change across photos. The system must focus on identity-related facial patterns. Apply the requirement and select the suitable method.

- A. Scene grouping
- B. Face grouping
- C. Text grouping
- D. Blur grouping

Correct Answer: B

Explanation:

Face grouping is suitable when the goal is to organize photos based on the same person's identity.

A surveillance camera records people entering an auditorium. Some faces are frontal, but some are partially covered by masks, spectacles, hair, or side pose. The system must still detect possible face regions before sending them to recognition. The detector is trained with varied appearances to handle these practical cases. Apply the scenario and select the main challenge.

- A. Pose-occlusion issue
- B. Stereo-depth issue
- C. File-format issue
- D. Text-layout issue

Correct Answer: A

Explanation:

Face detection in real scenes is affected by pose changes, occlusion, scale variation, lighting, and accessories.

A security system installed near a server room performs two stages. In the first stage, it draws a rectangular box around a visible face in the video frame. In the second stage, it compares that face with stored authorized profiles. The first stage finds the face location, and the second stage decides who the person is. Apply the concepts and select the correct interpretation.

- A. Recognition then detection
- B. Detection then recognition
- C. Segmentation then OCR
- D. Calibration then tracking

Correct Answer: B

Explanation:

Face detection locates the face. Face recognition identifies or verifies the person.

A face recognition module compares a query face with three stored profiles. The distance score with Profile A is 0.36, Profile B is 0.24, and Profile C is 0.31. In this system, a smaller score means the query face is more similar to the stored profile. The module must select the closest profile. Apply the rule and select the best match.

- A. Profile A
- B. Profile B
- C. Profile C
- D. Reject all

Correct Answer: B

Explanation:

The smallest distance score is 0.24, so Profile B is the closest match.

A college media team uploads photos from an FDP, a coding contest, a sports meet, and an alumni event. Some photos do not contain clear faces, but they show stage banners, laptops, certificates, playgrounds, or auditorium backgrounds. The system must still group them into correct event albums. It therefore uses event-related scene and context cues. Apply the concept and select the best cue.

- A. Face cue only
- B. File size cue
- C. Event-scene cue
- D. Camera brand cue

Correct Answer: C

Explanation:

Event-based photo grouping can use scene content, objects, banners, date, place, and visual similarity, not only faces.

A face recognition system works accurately when users face the camera directly under proper lighting. During evening hours, the same system gives wrong matches for side-facing faces and shadow-covered faces. The face detector still locates the face, but identity matching becomes unreliable. The developer wants to identify the likely cause. Apply the concept and select the likely reason.

- A. Storage format
- B. Album size
- C. Frame count
- D. Pose-light variation

Correct Answer: D

Explanation:

Face recognition can fail due to pose variation, illumination changes, occlusion, expression variation, and poor image quality.

1. Applying Eigenfaces in Face Recognition

A university security system stores face images of faculty members. Instead of comparing every pixel directly, the system represents each face using a smaller set of important face patterns learned from training images. These patterns capture major variations among faces and reduce computation during matching. The system uses this compact representation for recognition.

Apply the concept and select the suitable method.

- A. Eigenface method
- B. Hough method
- C. Stereo method
- D. Flow method

Correct Answer: A

Explanation:

Eigenfaces use PCA to represent face images using important face patterns. This reduces dimensionality and supports faster face recognition.

A face recognition system originally uses 500 pixel-based features for each face image. After applying the Eigenface method, only 80 important components are retained. The reduced representation is used for faster matching. The developer wants to know how many features were removed. Apply the feature reduction concept and select the removed feature count.

- A. 380
- B. 420
- C. 500
- D. 580

Correct Answer: B

Explanation:

$$\begin{aligned} \text{Removed Features} &= \text{Original Features} - \text{Retained Components} \\ \text{Removed Features} &= 500 - 80 = 420 \end{aligned}$$

So, 420 features are removed.

A face recognition system uses Eigenfaces for identity matching. Before extracting principal components, it creates an average face from all training face images. Each training face is then compared relative to this average face. This helps the system focus on meaningful variations among faces. Apply the Eigenface workflow and select the preprocessing step.

- A. Edge voting
- B. Depth scaling
- C. Mean face creation
- D. Motion filtering

Correct Answer: C

Explanation:

In the Eigenface method, the mean face is computed first. Then individual face differences from the mean are used for PCA-based representation.

A face recognition module compares a query face with three stored Eigenface representations. The distance score with Profile A is 0.42, Profile B is 0.27, and Profile C is 0.35. In this system, a lower distance score indicates a closer match. The system must select the best identity. Apply the matching rule and select the best match.

- A. Profile A
- B. Profile C
- C. Reject all
- D. Profile B

Correct Answer: D

Explanation:

The smallest distance score is 0.27, so Profile B is the closest match.

A face analysis system must locate facial landmarks such as eyes, nose, mouth, and jawline. The system must also adjust to changes in expression, such as smiling or opening the mouth. It uses both facial shape information and texture appearance to fit the face model accurately. This helps in facial alignment before recognition. Apply the concept and select the suitable model.

- A. Flow field model
- B. Appearance model
- C. Stereo depth model
- D. Hough line model

Correct Answer: B

Explanation:

Active Appearance Models combine shape and texture or appearance information. They are useful for facial landmark fitting and face alignment.

A face modeling system marks landmark points around the eyes, lips, nose, and face outline. It also analyzes the pixel texture inside the face region to understand facial appearance. The model needs both geometric landmark positions and intensity pattern information. This combined representation is used for face alignment. Apply the concept and select the information pair used.

- A. Shape and texture
- B. Depth and disparity
- C. Blur and focus
- D. Time and speed

Correct Answer: A

Explanation:

Active Appearance Models use shape information from landmarks and appearance or texture information from the image region.

A face recognition system is used at an airport. Passengers may look slightly left, right, up, or down while passing the camera. A simple 2D face model fails when pose changes are large. The system uses a model that represents the face surface structure in three dimensions. Apply the requirement and select the suitable model.

- A. Template image model
- B. Binary image model
- C. 3D shape model
- D. Text layout model

Correct Answer: C

Explanation:

3D shape models of faces help handle pose and viewpoint changes because they represent face structure in three dimensions.

A mobile face authentication system works well for frontal faces but fails when the user turns the head slightly. The developer wants to improve recognition by estimating the face structure and aligning it to a standard view before matching. The model should reduce the effect of head pose variation. Apply the concept and select the best use of a 3D face model.

- A. Pose alignment
- B. Audio filtering
- C. Edge thinning
- D. Text grouping

Correct Answer: A

Explanation:

A 3D face model can help estimate head pose and align the face to a standard view, improving recognition under pose variation.

A facial analysis system is used to track expressions in video lectures. The face shape changes when the person smiles, speaks, or raises eyebrows. The system must adjust facial landmarks while maintaining a valid face structure. It should model controlled changes in facial shape and appearance. Apply the scenario and select the suitable approach.

- A. Camera calibration
- B. Active appearance model
- C. Circle detection
- D. Stereo matching

Correct Answer: B

Explanation:

Active Appearance Models are useful for modeling facial shape and appearance variations caused by expression changes.

A researcher compares two face recognition systems. The first system uses PCA-based face patterns learned from 2D face images. The second system represents the face as a 3D surface to handle head rotation and pose variation. Both support recognition, but they represent faces differently. Apply the concepts and select the correct distinction.

- A. Eigenfaces use 3D pose only
- B. 3D models use pixel PCA only
- C. Both use only audio data
- D. Eigenfaces are 2D PCA-based; 3D models represent face surface shape

Correct Answer: D

Explanation:

Eigenfaces are based on PCA applied to 2D face images. 3D face models represent the geometric surface structure of the face and help handle pose variation.

A CCTV camera is installed at a college entrance to monitor student movement. The system first learns the usual background, such as the gate, wall, pathway, and notice board. When a person enters the scene, the system separates the moving person from the static background. The separated region is then passed to the tracking module. Apply the concept and select the suitable task.

- A. Foreground extraction
- B. Stereo calibration
- C. Shape projection
- D. Face alignment

Correct Answer: A

Explanation:

Foreground-background separation identifies moving or important foreground objects by separating them from the learned background scene.

A surveillance system monitors a parking area throughout the day. In the morning, the area is bright, but in the evening, the lighting gradually becomes dim. If the background model is not updated, shadows and brightness changes may be wrongly detected as moving objects. The developer updates the background model over time. Apply the concept and select the main purpose.

- A. Fixed object tracking
- B. Lighting adaptation
- C. Camera pose fitting
- D. Shape prior matching

Correct Answer: B

Explanation:

Updating the background model helps the system adapt to illumination changes, shadows, and slow background variations.

A security camera observes a laboratory corridor. The walls, doors, and floor remain mostly unchanged, but people walking through the corridor appear only in some frames. The system compares current frames with the learned background and marks changed regions as foreground. These regions are used for activity monitoring. Apply the scenario and select the detected region.

- A. Static wall region
- B. Door frame region
- C. Moving person region
- D. Floor tile region

Correct Answer: C

Explanation:

Foreground regions usually correspond to moving or newly appearing objects, such as people or vehicles, that differ from the background.

A CCTV camera monitors a garden pathway. The system wrongly detects moving tree leaves and their shadows as foreground objects. No person is present in the scene, but the background itself is changing due to wind and sunlight. The engineer must identify the practical challenge. Apply the scenario and select the issue.

- A. Stable background
- B. Fixed illumination
- C. Static foreground
- D. Dynamic background

Correct Answer: D

Explanation:

Dynamic background occurs when background elements such as leaves, water, shadows, or lighting vary over time and cause false foreground detection.

A surveillance system produces a binary foreground mask for a corridor video. White pixels represent moving foreground regions, and black pixels represent background regions. In one frame, a walking person appears as a connected white region in the mask. The system uses this region for tracking. Apply the mask concept and select the white region meaning.

- A. Moving object area
- B. Camera noise only
- C. Background surface
- D. Image border area

Correct Answer: A

Explanation:

In a typical foreground mask, white pixels represent detected foreground objects, while black pixels represent background.

A surveillance system tracks a person walking through a crowded hallway. The person's movement is not perfectly predictable because of turns, speed changes, and partial occlusion. The tracker maintains several possible positions of the person and updates them using new observations from each frame. This helps the system continue tracking under uncertainty. Apply the concept and select the suitable method.

- A. Hough voting
- B. Particle filtering
- C. Eigenface matching
- D. Depth defocus

Correct Answer: B

Explanation:

Particle filters track objects under uncertainty by maintaining multiple possible states, called particles, and updating them using observations.

A particle-filter-based tracker follows a moving student in a campus video. Before observing the next frame, the tracker predicts possible new positions based on the student's previous motion. These predicted positions are represented by particles. Later, the particles are updated using the actual image observation. Apply the concept and select the stage described.

- A. Prediction stage
- B. Recognition stage
- C. Calibration stage
- D. Quantization stage

Correct Answer: A

Explanation:

The prediction stage estimates possible future object states using motion information before checking the next observation.

A camera tracks a pedestrian crossing the road. After predicting several possible positions, the tracker compares each particle with the current image observation. Particles closer to the actual pedestrian receive higher importance, while unlikely particles receive lower importance. The tracker then focuses on the more reliable particles. Apply the concept and select the stage described.

- A. Image sampling
- B. Feature pooling
- C. Weight update
- D. Shape coding

Correct Answer: C

Explanation:

In particle filtering, the update step assigns weights to particles based on how well they match the current observation.

A person being tracked in a railway station is temporarily hidden behind another passenger. A simple tracker immediately loses the target. A particle filter maintains multiple possible target locations during the occlusion and can recover when the person reappears. This improves tracking continuity. Apply the scenario and select the main benefit.

- A. Better color storage
- B. Better line detection
- C. Better face cropping
- D. Better occlusion handling

Correct Answer: D

Explanation:

Particle filters are useful in occlusion because they maintain multiple hypotheses about the object location until reliable observations return.

A surveillance system first separates moving objects from the background in each video frame. Then, for a selected moving person, it uses a particle filter to estimate the person's position over time. The first stage detects possible moving regions, and the second stage maintains the target's motion path. This combined system supports continuous monitoring. Apply the workflow and select the correct order.

- A. Particle tracking → foreground separation
- B. Foreground separation → particle tracking
- C. Face recognition → background update
- D. Shape matching → camera calibration

Correct Answer: B

Explanation:

Foreground-background separation first identifies moving objects. Particle filtering can then track the selected object across frames.

A factory vision system checks whether a metal washer has the correct circular shape. The system first extracts the edge map of the captured image. It then compares the washer's edge pattern with a stored circular template. The object is accepted when the template edge closely matches the image edge. Apply the concept and select the suitable method.

- A. Chamfer matching
- B. Stereo matching
- C. Gait modeling
- D. Pose tracking

Correct Answer: A

Explanation:

Chamfer matching is used for edge-based shape or template matching. It compares a template shape with edge information in an image.

A road sign detection system uses chamfer matching to compare a triangular sign template with an edge image. Four template edge points have nearest image-edge distances of 2, 4, 3, and 1 pixels. A lower total distance indicates a better shape match. The system calculates the total matching cost. Apply the matching cost concept and select the total cost.

- A. 8
- B. 10
- C. 12
- D. 14

Correct Answer: B

Explanation:

$$Total\ Cost = 2 + 4 + 3 + 1 = 10$$

So, the chamfer matching cost is 10.

A vision system must identify whether a detected object is a circle, triangle, or rectangle. The chamfer matching cost with the circle template is 18, triangle template is 9, and rectangle template is 14. The system selects the template with the lowest cost as the best match. Apply the matching rule and select the best template.

- A. Circle template
- B. Rectangle template
- C. Triangle template
- D. Reject all templates

Correct Answer: C

Explanation:

In chamfer matching, the lowest cost indicates the closest shape match. The triangle template has the lowest cost, 9, so it is selected.

A traffic camera detects a road sign, but part of the sign boundary is missing due to shadow. The system still tries to compare the available edge pattern with stored templates. A method that uses edge distances can tolerate small boundary gaps better than exact pixel-by-pixel matching. The system selects the closest template based on edge distance. Apply the scenario and select the suitable approach.

- A. Exact pixel copying
- B. Audio pattern coding
- C. Camera pose fitting
- D. Chamfer-based matching

Correct Answer: D

Explanation:

Chamfer matching compares edge-template distances and can be useful when boundaries are slightly broken or incomplete.

A surveillance camera monitors a corridor in a hospital. A person is detected in one frame and appears at a slightly different position in the next frame. The system links these detections to maintain the same identity over time. This helps security staff follow the person's movement path. Apply the concept and select the suitable task.

- A. Object tracking
- B. Image sampling
- C. Face cropping
- D. Scene labeling

Correct Answer: A

Explanation:

Object tracking follows the same object across consecutive frames and maintains its identity over time.

A shopping mall camera tracks a customer walking through a crowded area. The customer temporarily disappears behind a pillar and then reappears. A weak tracker assigns a new ID after reappearance, treating the same customer as a different person. The system fails to maintain continuous identity. Apply the scenario and select the main problem.

- A. Blur estimation
- B. Identity switch
- C. Template scaling
- D. Color sampling

Correct Answer: B

Explanation:

When a tracker gives a new identity to the same object after occlusion, it is called an identity switch.

A railway station surveillance system tracks passengers across video frames. When a passenger is hidden behind another person for a few frames, the tracker predicts the likely location using previous motion information. When the passenger reappears, the system reconnects the detection to the same track. Apply the concept and select the suitable handling strategy.

- A. Track prediction
- B. Edge thinning
- C. Image binning
- D. Color inversion

Correct Answer: A

Explanation:

Track prediction helps estimate the likely position of a temporarily hidden object and supports recovery after occlusion.

A traffic monitoring system detects vehicles in every video frame using an object detector. After detection, it links vehicle boxes across frames based on position, size, and appearance similarity. This allows the system to count vehicles and estimate their movement paths. Apply the workflow and select the correct approach.

- A. Detection-free coding
- B. Segmentation-only flow
- C. Tracking-by-detection
- D. Calibration-by-depth

Correct Answer: C

Explanation:

Tracking-by-detection first detects objects in each frame and then associates detections across frames to form tracks.

A camera records two people crossing each other in a corridor. For a few frames, their bounding boxes overlap strongly. After they separate, the tracker must decide which person is which by using appearance and motion history. Without this handling, the tracker may swap their identities. Apply the scenario and select the main challenge.

- A. Occlusion handling
- B. Blur sharpening
- C. Depth focusing
- D. Image sampling

Correct Answer: A

Explanation:

When objects overlap or hide each other, tracking systems must handle occlusion to avoid identity loss or identity swapping.

A robot tracks a tool moving on a workbench. The tool is partly visible in some frames, so the robot uses the tool's edge template to confirm its shape when appearance is unclear. It then links the confirmed tool location across frames. The system combines edge-based shape matching with temporal tracking. Apply the workflow and select the correct combination.

- A. PCA and sampling
- B. OCR and calibration
- C. HMM and quantization
- D. Chamfer matching and tracking

Correct Answer: D

Explanation:

Chamfer matching helps confirm the object shape using edge-template matching, while tracking links the object across frames over time.

A college campus uses cameras at the entrance gate, corridor, and parking area. A person first appears in the entrance camera and later appears in the corridor camera. The system must connect these observations to understand the person's movement path across different camera views. This helps security staff monitor the movement continuously. Apply the concept and select the suitable task.

- A. Multi-camera tracking
- B. Face pose fitting
- C. Image noise removal
- D. Road sign labeling

Correct Answer: A

Explanation:

Multi-camera tracking combines information from different cameras to follow the same object or person across multiple views.

A surveillance system monitors a large auditorium using several cameras. One camera captures the front stage, another covers the audience, and another covers the entrance. Individually, each camera gives only partial coverage. By combining views, the system obtains a more complete understanding of the scene. Apply the scenario and select the main benefit.

- A. Smaller image size
- B. Wider scene coverage
- C. Lower frame count
- D. Fixed object color

Correct Answer: B

Explanation:

Combining multiple camera views improves coverage and reduces blind spots in large surveillance areas.

A shopping mall surveillance system tracks a person moving from Camera 1 to Camera 2. The person disappears from the first camera and reappears in another view after a few seconds. The system compares clothing color, body shape, walking pattern, and time gap to decide whether it is the same person. This helps maintain identity across cameras. Apply the concept and select the suitable process.

- A. Lens distortion correction
- B. Edge thinning
- C. Cross-camera matching
- D. Background smoothing

Correct Answer: C

Explanation:

Cross-camera matching links the same person or object across different cameras using appearance, time, location, and motion cues.

A sports arena uses multiple cameras to follow players. The cameras capture the same player from different angles. To combine their views correctly, the system must know the geometric relationship between the cameras. This helps map observations from one camera view to another. Apply the concept and select the required step.

- A. Texture filtering
- B. Face grouping
- C. Audio matching
- D. Camera calibration

Correct Answer: D

Explanation:

Camera calibration helps estimate the geometric relationship between cameras, which is important for combining multiple views correctly.

Two security cameras are installed in a corridor with overlapping fields of view. A person appears in the overlap region, so both cameras capture the person at nearly the same time. The system uses this shared region to confirm that the detections in both cameras belong to the same person. This improves tracking reliability. Apply the concept and select the useful cue.

- A. Overlap region
- B. File extension
- C. Pixel sampling
- D. Audio channel

Correct Answer: A

Explanation:

An overlap region allows multiple cameras to observe the same object, making identity matching and view fusion more reliable.

A security system identifies a person from a distant camera view where the face is not clear. The system observes the person's walking style, including step length, body movement, leg swing, and walking rhythm. These movement patterns are compared with stored walking patterns. This supports identification even without a clear face. Apply the concept and select the suitable biometric method.

- A. Iris matching
- B. Gait analysis
- C. Fingerprint scan
- D. Voice coding

Correct Answer: B

Explanation:

Human gait analysis identifies or studies a person using walking patterns and body movement over time.

A gait recognition system records a person walking across a corridor. The system extracts stride length, walking speed, leg movement pattern, and body silhouette changes. These features are used to compare the walking style with stored gait data. The system avoids using face features because the camera view is distant. Apply the scenario and select the relevant feature group.

- A. Walking pattern cues
- B. Camera lens cues
- C. Text layout cues
- D. Road marking cues

Correct Answer: A

Explanation:

Gait analysis uses walking-related features such as stride, step rhythm, silhouette sequence, posture, and body motion.

A surveillance camera captures a person walking at night. The face is blurred due to distance and poor lighting, but the full body movement is visible across frames. The system must still support identification using motion-based body features. The best method should not rely mainly on facial texture. Apply the requirement and select the suitable approach.

- A. Face texture matching
- B. Roadway detection
- C. Gait-based recognition
- D. Static OCR reading

Correct Answer: C

Explanation:

Gait-based recognition is useful when the face is unclear but walking pattern and body movement are visible.

A human gait analysis system studies a walking sequence. It observes repeated movement of the legs, arms, and body posture over time. The system uses the periodic nature of walking to form a gait signature. This signature is then used for person identification or activity analysis. Apply the concept and select what is mainly analyzed.

- A. Repeated walking cycle
- B. Static face color
- C. Camera file name
- D. Document layout

Correct Answer: A

Explanation:

Gait analysis studies the repeated walking cycle, including body posture and limb movement over time.

A railway station uses multiple cameras to observe passengers from different directions. One camera captures a side view, another captures a front view, and another captures a rear view. The system combines these views to obtain a more reliable walking pattern when one view is blocked or unclear. This improves gait-based identification. Apply the workflow and select the correct combination.

- A. OCR and sampling
- B. Blur and defocus
- C. Pose and lighting
- D. Multi-view gait analysis

Correct Answer: D

Explanation:

Multi-view gait analysis combines walking information from different camera views to improve identification when one view is incomplete or occluded.

An autonomous car uses a front-facing camera while moving on a highway. The system must understand road boundaries, lane markings, traffic signs, nearby vehicles, and pedestrians. This visual information helps the vehicle decide whether to stay in lane, slow down, or stop. The camera-based module acts as an important part of the vehicle perception system. Apply the concept and select the suitable application.

- A. Vehicle vision system
- B. Medical scan system
- C. Document OCR system
- D. Album grouping system

Correct Answer: A

Explanation:

An in-vehicle vision system uses camera input to understand roadways, lanes, signs, vehicles, pedestrians, and traffic situations.

A self-driving vehicle travels on a road where the lane markings are partly faded. The system still needs to identify the drivable road region using road texture, color, boundary cues, and scene layout. It separates the road surface from sidewalks, buildings, and vegetation. This helps the vehicle remain within the safe driving area. Apply the scenario and select the suitable task.

- A. Face landmarking
- B. Roadway locating
- C. Tumor outlining
- D. Logo matching

Correct Answer: B

Explanation:

Roadway locating identifies the drivable road region by using visual cues such as texture, color, boundaries, and scene structure.

A camera mounted on a car captures white and yellow lane markings on the road. The system enhances the image, detects strong line-like structures, and uses the detected markings to estimate the lane position. This output helps the vehicle maintain its lane and avoid drifting. The task focuses mainly on painted road lines. Apply the concept and select the correct task.

- A. Face matching
- B. Gait tracking
- C. Lane marking detection
- D. Background modeling

Correct Answer: C

Explanation:

Road marking detection identifies painted lane boundaries, arrows, zebra crossings, and other road surface markings.

A vehicle camera captures a red circular sign near the roadside. The system first detects the sign region, then analyzes its shape, color, and internal symbol. Finally, it classifies it as a speed-limit sign. The output is used to assist driving decisions. Apply the workflow and select the correct task.

- A. Road texture grouping
- B. Pedestrian masking
- C. Lane border fitting
- D. Traffic sign recognition

Correct Answer: D

Explanation:

Road sign identification detects and classifies traffic signs using visual features such as shape, color, symbol, and text.

An in-vehicle camera observes long straight lane boundaries on a highway. After edge detection, the system must find dominant straight lines corresponding to the lane markings. The method should handle broken lane markings by grouping edge points that support the same line. This helps estimate the vehicle's lane position. Apply the requirement and select the suitable method.

- A. Hough line detection
- B. Face recognition
- C. Depth defocus
- D. Gait analysis

Correct Answer: A

Explanation:

Hough Transform is commonly used to detect line-like road markings by allowing edge points to vote for possible line parameters.

A driver assistance system identifies road signs from live camera frames. For a warning sign, the system uses triangular shape, red border, and internal symbol pattern. For speed-limit signs, it uses circular shape, red border, and number information. These features help separate one type of sign from another. Apply the concept and select the useful feature set.

- A. Lens and focus
- B. Shape and symbol
- C. Frame and speed
- D. Pose and gait

Correct Answer: B

Explanation:

Road sign recognition commonly uses shape, color, border, symbol, and text/number features.

A vehicle vision system has two modules. The first module identifies the general drivable surface ahead of the car. The second module detects white and yellow painted lines that divide lanes. Both modules support lane keeping, but they focus on different visual information. Apply the concepts and select the correct interpretation.

- A. Sign detection finds road texture
- B. Roadway detection finds faces
- C. Roadway locates road area
- D. Road marking detects organs

Correct Answer: C

Explanation:

Roadway detection locates the drivable road surface, while road marking detection locates painted lane lines and symbols on the road.

A vehicle travels during evening time, and the camera captures a traffic sign with low contrast. The sign is still partly visible because of its shape and reflective border, but the internal symbol is unclear. The system must first detect the probable sign region before final classification. The detection stage should focus on visual cues that remain visible. Apply the scenario and select the best cue.

- A. Audio cue only
- B. Shape-color cue
- C. File-size cue
- D. Battery cue only

Correct Answer: B

Explanation:

Road sign detection can use shape and color cues, especially when symbol details are weak due to poor lighting or distance.

A car-mounted camera continuously observes lane markings. The detected left and right lane boundaries show that the vehicle is moving closer to the right lane boundary. The system uses this information to warn the driver before the vehicle crosses the lane. This is part of driver assistance. Apply the concept and select the suitable use.

- A. Lane departure warning
- B. Face identity grouping
- C. Medical boundary fitting
- D. Album event sorting

Correct Answer: A

Explanation:

Lane departure warning systems use road marking detection to determine whether the vehicle is drifting out of its lane.

An autonomous vehicle receives video from a front camera. The system first locates the drivable road region, then detects lane markings, and finally identifies road signs near the roadside. These outputs are combined to support safe path planning and driving decisions. The workflow is used for road scene understanding. Apply the workflow and select the correct order.

- A. Sign recognition → album grouping → face detection
- B. Roadway locating → marking detection → sign identification
- C. Medical segmentation → gait matching → road detection
- D. Face detection → OCR reading → pose estimation

Correct Answer: B

Explanation:

A typical in-vehicle vision workflow may include roadway detection, road marking detection, and road sign recognition for driving assistance and perception.

An autonomous car moves near a school zone during morning hours. The camera captures children standing near the zebra crossing and some adults walking along the roadside. The system must locate human regions before predicting whether they may enter the road. This step helps the vehicle slow down safely. Apply the concept and select the suitable task.

- A. Pedestrian detection
- B. Road texture fitting
- C. Album face grouping
- D. Tumor segmentation

Correct Answer: A

Explanation:

Pedestrian detection locates human regions in road scenes, usually by drawing bounding boxes around pedestrians.

A vehicle-mounted camera observes a crowded street. The system identifies multiple pedestrians at different positions in the frame. For each pedestrian, it marks the location using rectangular regions so that the tracking module can follow them across frames. These marked regions are also used for collision risk estimation. Apply the concept and select the expected output.

- A. Camera pose values
- B. Pedestrian boxes
- C. Road sign labels
- D. Lane curve points

Correct Answer: B

Explanation:

Pedestrian localization commonly outputs bounding boxes around detected pedestrians.

A driver assistance system detects people walking near a bus stop. Some pedestrians wear different colored clothes, and the background contains vehicles, trees, and signboards. The system relies on human-like shape, vertical body structure, motion pattern, and learned visual features to locate pedestrians. These cues help separate people from other roadside objects. Apply the scenario and select the useful cue group.

- A. File size cues
- B. Camera brand cues
- C. Human shape cues
- D. Audio pitch cues

Correct Answer: C

Explanation:

Pedestrian detection uses human shape, body structure, motion, appearance, and learned visual features.

An autonomous vehicle approaches a parked bus. A pedestrian is partly visible from behind the bus, with only the upper body and head clearly seen. The system must still identify the possible pedestrian region and treat it as a safety risk. This requires handling partial visibility in real-time road scenes. Apply the scenario and select the main challenge.

- A. Clear lane marking
- B. Stable background
- C. Fixed camera view
- D. Partial occlusion

Correct Answer: D

Explanation:

Partial occlusion occurs when a pedestrian is partly hidden by another object, such as a vehicle, pole, or crowd.

A pedestrian is detected near a zebra crossing. In consecutive frames, the pedestrian's position moves closer to the vehicle path. The autonomous system combines pedestrian location and motion direction to decide whether braking may be required. The decision depends on both detection and motion understanding. Apply the concept and select the suitable use.

- A. Risk prediction
- B. Face grouping
- C. Texture coding
- D. Lens correction

Correct Answer: A

Explanation:

Autonomous vehicles use pedestrian location and motion cues to estimate collision risk and support braking decisions.

A self-driving car receives continuous camera input while driving in city traffic. The system detects lanes, pedestrians, vehicles, traffic lights, road signs, and obstacles. It combines these outputs to understand the road scene and plan safe movement. This complete visual understanding supports autonomous navigation. Apply the concept and select the suitable system.

- A. Document analysis
- B. Vehicle perception
- C. Medical diagnosis
- D. Album sorting

Correct Answer: B

Explanation:

Autonomous vehicle perception combines multiple visual tasks such as lane detection, object detection, sign recognition, and pedestrian localization.

A vehicle perception system processes a front-camera video. First, it locates the drivable road area. Next, it detects lane markings and road signs. It then locates pedestrians and nearby vehicles. Finally, the planning module uses this information to support safe driving decisions. Apply the workflow and select the correct order.

- A. Face match → OCR → gait cue
- B. Tumor map → depth cue → album cue
- C. Road scene detection → object localization → decision support
- D. File sorting → image naming → audio coding

Correct Answer: C

Explanation:

A vehicle perception workflow extracts road scene information, localizes important objects, and passes the result to decision or planning modules.

An autonomous vehicle uses a camera, LiDAR, and radar in a busy road environment. The camera provides visual details such as signs and lane markings, LiDAR gives accurate depth, and radar helps detect moving objects in poor visibility. The system combines these sensor outputs for robust perception. This helps overcome the limitations of a single sensor. Apply the concept and select the suitable approach.

- A. Single-image OCR
- B. Face-only matching
- C. Lane-only grouping
- D. Sensor fusion

Correct Answer: D

Explanation:

Sensor fusion combines data from multiple sensors to improve reliability in autonomous vehicle perception.

A vehicle camera detects pedestrians accurately in daytime. At night, the detector misses some pedestrians wearing dark clothing near poorly lit roads. The system still detects bright road signs and lane markings, but human regions become difficult to separate from the background. The developer must improve the detector for low-light cases. Apply the scenario and select the likely issue.

- A. Low-light visibility
- B. High file storage
- C. Clear object boundary
- D. Fixed road texture

Correct Answer: A

Explanation:

Poor lighting and low contrast can reduce pedestrian detection accuracy, especially when clothing blends with the background.

A child suddenly appears near the road edge, and the perception system detects the child as a pedestrian. The tracking module estimates that the child is moving toward the vehicle path. The vehicle uses this information to activate a warning and prepare braking. The system response depends on perception and prediction. Apply the concept and select the suitable function.

- A. Album clustering
- B. Road sign printing
- C. Emergency braking
- D. Camera formatting

Correct Answer: C

Explanation:

Autonomous emergency braking uses pedestrian detection, tracking, and risk prediction to reduce collision chances.

A hospital uses MRI scans to study brain tumor cases. The radiologist wants the computer system to separate the tumor region from normal brain tissues. The extracted tumor region will be used for area measurement, treatment planning, and follow-up comparison. The system must mark only the meaningful abnormal region in the scan. Apply the concept and select the suitable task.

- A. Tumor segmentation
- B. Roadway detection
- C. Album grouping
- D. Face matching

Correct Answer: A

Explanation:

Medical image segmentation separates meaningful medical regions such as tumors, organs, lesions, or tissues from the rest of the scan.

A CT scan contains liver, kidney, bone, and surrounding soft tissue regions. Before measuring the liver volume, the system must isolate only the liver area from the entire scan. The output is later used by doctors to examine organ size and structure. The task focuses on separating one anatomical region from others. Apply the scenario and select the suitable output.

- A. Camera pose
- B. Liver region
- C. Road sign
- D. Face box

Correct Answer: B

Explanation:

Organ segmentation extracts a specific anatomical structure, such as the liver, kidney, heart, or brain, from a medical image.

A medical image system processes an ultrasound image. After segmentation, the tumor area is shown in white, while the surrounding normal tissue is shown in black. The white region is used for further analysis and measurement. The doctor checks whether the marked region matches the actual tumor. Apply the concept and select what the white region represents.

- A. Image noise
- B. Camera lens
- C. Target lesion
- D. File border

Correct Answer: C

Explanation:

In a binary segmentation mask, the highlighted or white region often represents the target object or region of interest.

A medical segmentation system identifies a lesion region in a skin image. The segmented lesion contains 600 pixels, and each pixel represents 0.5 mm^2 of real tissue area. The doctor wants to estimate the lesion area from the segmentation result. This helps compare lesion size during follow-up. Apply the measurement concept and select the lesion area.

- A. 100 mm^2
- B. 200 mm^2
- C. 250 mm^2
- D. 300 mm^2

Correct Answer: D

Explanation:

$$\begin{aligned} \text{Area} &= \text{Number of Pixels} \times \text{Area per Pixel} \\ \text{Area} &= 600 \times 0.5 = 300 \text{ mm}^2 \end{aligned}$$

So, the lesion area is 300 mm^2 .

A medical image system segments blood vessels from retinal images. Some vessels are thin, low-contrast, and close to noisy background regions. If the system uses only simple thresholding, it misses many weak vessel structures. The developer decides to use a more robust segmentation approach. Apply the scenario and select the main challenge.

- A. Weak boundaries
- B. Clear labels

- C. Fixed lighting
- D. Strong texture

Correct Answer: A

Explanation:

Medical segmentation often faces weak boundaries, noise, low contrast, overlapping tissues, and intensity similarity between regions.

A medical training app uses a tablet camera to view a physical mannequin. When the camera points at the mannequin, the app overlays digital labels for heart, lungs, and ribs on the real-world view. The trainee can still see the actual mannequin while digital guidance is added on top. This supports interactive learning. Apply the concept and select the suitable technology.

- A. Virtual reality
- B. Augmented reality
- C. Stereo calibration
- D. Image compression

Correct Answer: B

Explanation:

Augmented Reality overlays digital information on the real-world view while keeping the real environment visible.

A mobile AR app places a virtual chair on the floor through the phone camera. As the user moves the phone, the virtual chair must remain fixed at the same real-world location. The system tracks the camera pose and the floor surface continuously. This prevents the object from sliding unrealistically. Apply the scenario and select the required function.

- A. Audio filtering
- B. Text binarizing
- C. Pose tracking
- D. Face grouping

Correct Answer: C

Explanation:

AR systems require pose tracking to align virtual objects with the real-world scene as the camera moves.

A mixed reality headset displays a virtual machine model inside a real laboratory. The user can walk around it, and the virtual model appears to stay on the real table. The user can also interact with the model, rotate its parts, and view internal components. Real and virtual objects are spatially connected. Apply the concept and select the suitable technology.

- A. OCR system
- B. Gait analysis

- C. Lane tracking
- D. Mixed reality

Correct Answer: D

Explanation:

Mixed Reality allows real and virtual objects to coexist and interact more naturally in a shared space.

A mobile app simply overlays arrow directions on a real street view. A headset-based system places a virtual robot on a real floor and allows the user to walk around and interact with it. Both use real-world views, but the second system supports stronger interaction between physical and virtual objects. Apply the concepts and select the correct distinction.

- A. AR overlay; MR interaction
- B. MR overlay; AR isolation
- C. AR sampling; MR filtering
- D. MR OCR; AR tracking

Correct Answer: A

Explanation:

AR mainly overlays digital content on the real world. MR supports deeper interaction and spatial relationship between real and virtual objects.

A surgeon uses a camera-based guidance system during a procedure. The system overlays the planned incision path and important anatomical landmarks on the live view of the patient. The overlay must align correctly with the real body surface to avoid clinical errors. Accurate tracking and registration are essential. Apply the scenario and select the most important requirement.

- A. Album naming
- B. Road marking
- C. Overlay alignment
- D. Face grouping

Correct Answer: C

Explanation:

In medical AR, accurate alignment or registration between virtual guidance and the real patient view is essential for safe and meaningful use.

A university gallery app automatically groups photos from seminars, lab sessions, sports events, and cultural functions. It performs well when clear faces are available, but it wrongly groups some event photos when faces are absent. For example, lab photos and seminar photos with similar lighting are placed in the same album. The developer wants to improve grouping by using more contextual cues. Analyze the issue and select the best improvement.

- A. Add scene-event cues
- B. Add camera battery data

- C. Add file size sorting
- D. Add grayscale conversion

Correct Answer: A

Reason:

Photo album organization should not depend only on faces. Scene objects, banners, location, time, and visual context help group event photos more accurately.

A security system at a laboratory entrance first draws a box around a person's face. Later, another module checks whether the detected face belongs to an authorized staff member. During testing, the system correctly draws the face box but gives the wrong identity. The technician says that face detection has failed. Analyze the statement and select the correct interpretation.

- A. Detection failed first
- B. Recognition failed later
- C. Calibration failed first
- D. Segmentation failed later

Correct Answer: B

Reason:

Face detection locates the face region. Face recognition identifies or verifies the person. Since the face box was correct but identity was wrong, recognition failed.

3. Analyze Eigenface Limitation

A face recognition system uses Eigenfaces for staff identification. It performs well when images are frontal and captured under similar lighting. However, its accuracy decreases when faces are captured with side poses, shadows, or strong expression changes. The system represents faces using PCA-based 2D appearance patterns. Analyze the limitation and select the best reason.

- A. Eigenfaces ignore all pixels
- B. Eigenfaces detect only lanes
- C. Eigenfaces are pose-light sensitive
- D. Eigenfaces use depth maps only

Correct Answer: C

Reason:

Eigenfaces are based on 2D appearance variation. They can be affected by pose, illumination, expression, and alignment changes.

An airport face recognition system must identify passengers who may not always look directly at the camera. A 2D face model gives poor results when the head is turned. The system designer wants a model that can represent facial surface structure and support pose correction. The goal is to reduce recognition failure due to viewpoint change. Analyze the requirement and select the suitable model.

- A. 2D edge template
- B. 3D face model

- C. Binary text model
- D. Audio gait model

Correct Answer: B

Reason:

3D face models represent the surface structure of the face and are useful for handling pose and viewpoint variations.

A CCTV system monitors a garden pathway. It correctly detects people during normal conditions. On windy afternoons, tree leaves and their shadows are repeatedly detected as moving foreground objects even when no person is present. The background itself is changing over time. Analyze the cause and select the main issue.

- A. Dynamic background
- B. Stable foreground
- C. Fixed illumination
- D. Clear occlusion

Correct Answer: A

Reason:

Moving leaves, water, shadows, and illumination changes create dynamic background problems. These may be wrongly detected as foreground.

A surveillance camera tracks a person moving through a crowded railway station. For a few frames, the person is partially hidden by other passengers. A simple tracker loses the person immediately, while another tracker maintains several possible locations and recovers when the person reappears. The second tracker handles uncertainty better. Analyze the method and select the correct explanation.

- A. It stores only edges
- B. It uses multiple hypotheses
- C. It applies OCR matching
- D. It removes all motion

Correct Answer: B

Reason:

Particle filters maintain multiple possible states or hypotheses. This helps object tracking under uncertainty, noise, and partial occlusion.

A robot must recognize a tool on a workbench using its edge shape. The image is slightly noisy, and part of the tool boundary is broken due to shadow. Exact pixel-by-pixel template matching fails because the edge map is incomplete. The system needs a shape matching method that compares template edges with nearby image edges. Analyze the requirement and select the suitable method.

- A. Depth matching
- B. Color pooling
- C. Chamfer matching
- D. Gait matching

Correct Answer: C

Reason:

Chamfer matching compares template edge points with nearby image edges using distance. It is useful for edge-based shape matching, especially when boundaries are slightly incomplete.

A campus surveillance system has cameras at the gate, corridor, and parking area. A person disappears from the gate camera and appears later in the corridor camera. The system must decide whether both detections belong to the same person using appearance, movement direction, time gap, and camera layout. This helps maintain continuous monitoring. Analyze the task and select the correct concept.

- A. Single-frame detection
- B. Medical segmentation
- C. Multi-camera matching
- D. Road sign grouping

Correct Answer: C

Reason:

Multi-camera matching links detections of the same object or person across different camera views using appearance, time, location, and motion cues.

An autonomous vehicle approaches a parked bus near a school. A child is partially visible behind the bus and starts moving toward the road. The perception system must treat this partial human region as a possible pedestrian and prepare for braking. Missing this detection may cause a safety risk. Analyze the scenario and select the key challenge.

- A. Partial occlusion
- B. Album grouping
- C. Face alignment
- D. Organ labeling

Correct Answer: A

Reason:

Pedestrians may be partially hidden by vehicles, poles, or crowds. Handling partial occlusion is critical for autonomous vehicle safety.

A mobile navigation app overlays arrows on the live street view to guide the user. A headset-based training system places a virtual machine on a real table and lets the user walk around it, inspect parts, and interact with them. Both use the real environment, but the second system supports stronger spatial interaction. Analyze the two systems and select the correct distinction.

- A. AR uses no camera; MR uses text
- B. AR overlays; MR enables interaction
- C. AR replaces reality; MR removes objects
- D. AR uses OCR; MR uses face boxes

Correct Answer: B

Reason:

Augmented Reality mainly overlays digital content on the real-world view. Mixed Reality allows stronger interaction between real and virtual objects in a shared space.

A photo album app groups images of the same person across different events. It works well when the face is clear, but fails when the person appears in profile view or under poor lighting. Some photos are grouped based on similar background instead of identity. The developer wants to reduce wrong person-wise grouping. Analyze the issue and select the best correction.

- A. Improve face feature matching
- B. Use only image file size
- C. Sort photos by brightness
- D. Remove all event metadata

Correct Answer: A

Reason:

Person-wise grouping depends mainly on reliable facial feature matching. Pose and lighting variation can reduce recognition accuracy.

A facial expression analysis system tracks landmarks around the eyes, mouth, nose, and jawline. When a person smiles or speaks, the landmark positions and facial texture change together. A simple rigid face template cannot adjust properly to these changes. The system needs a model that handles both shape and appearance variation. Analyze the requirement and select the suitable model.

- A. Stereo depth model
- B. Active appearance model
- C. Hough circle model
- D. Background model

Correct Answer: B

Reason:

Active Appearance Models use both facial shape landmarks and texture information, making them suitable for expression and appearance changes.

A hospital corridor surveillance system detects a nurse correctly in each frame. However, when another person crosses in front, the tracker loses the nurse's identity and continues tracking the wrong person. The bounding boxes are detected, but identity association between frames fails. The security team wants to identify the main tracking problem. Analyze the situation and select the correct issue.

- A. Camera calibration
- B. Image quantization
- C. Identity switching
- D. Road segmentation

Correct Answer: C

Reason:

Identity switching occurs when a tracker assigns the tracked identity to another object, often during occlusion or object overlap.

A surveillance system produces a foreground mask where moving people should appear white and background should appear black. During testing, a large wall region is also marked white whenever sunlight changes suddenly. The system is detecting illumination variation as foreground. This creates false alarms. Analyze the cause and select the best explanation.

- A. Illumination change
- B. Fixed camera pose
- C. Stable background
- D. Clear object boundary

Correct Answer: A

Reason:

Sudden lighting changes can make background pixels appear different from the learned model, causing false foreground detection.

A security camera is placed far from a building entrance. The captured faces are too small for reliable recognition, but the full body movement is visible over several frames. The system compares walking rhythm, stride, posture, and silhouette motion. It uses this information for identity support. Analyze the scenario and select the suitable method.

- A. Face cropping
- B. Road detection
- C. Gait analysis
- D. Text matching

Correct Answer: C

Reason:

Gait analysis is useful when the face is unclear but walking pattern and body motion are visible across frames.

An in-vehicle vision system detects a circular red-bordered sign correctly. However, it misclassifies the sign because the number inside the sign is blurred due to motion. The sign region is localized, but the internal symbol recognition is unreliable. The developer wants to identify the failed stage. Analyze the situation and select the failed component.

- A. Roadway locating
- B. Sign classification
- C. Lane detection
- D. Gait modeling

Correct Answer: B

Reason:

The sign was detected correctly, but the internal number or symbol was misread. Therefore, the classification stage failed.

An autonomous vehicle detects lane markings, road signs, pedestrians, and nearby vehicles. In one case, the lane is detected correctly, but the system fails to slow down because a pedestrian near the crossing is missed. The planning module receives incomplete perception information. The safety issue occurs before decision making. Analyze the issue and select the main failure.

- A. Pedestrian perception
- B. Album clustering
- C. Face recognition
- D. MRI segmentation

Correct Answer: A

Reason:

The vehicle failed because the perception module missed a pedestrian. This affects the downstream planning and braking decision.

A tumor segmentation system marks most of the tumor region correctly, but it also includes nearby healthy tissue in the segmented mask. The doctor says the result is useful but needs correction before measurement. The system has over-expanded the abnormal region. This can affect area estimation. Analyze the output and select the correct issue.

- A. Under-segmentation
- B. Over-segmentation
- C. Face detection
- D. Road tracking

Correct Answer: B

Reason:

Over-segmentation occurs when the segmented region includes extra non-target areas, such as healthy tissue around a tumor.

A medical AR system overlays a planned incision line on a patient's live camera view. When the camera moves, the virtual line shifts away from the correct body location. The overlay is visible, but it is not spatially aligned with the real anatomy. The system needs better tracking and registration. Analyze the problem and select the main issue.

- A. Album grouping
- B. Gait mismatch
- C. Overlay misalignment
- D. Sign misclassification

Correct Answer: C

Reason:

AR systems require accurate registration and tracking so digital overlays remain aligned with real-world objects.

A training lab wants students to place a virtual engine model on a real table and interact with its parts. The model should stay fixed on the table, respond to user actions, and appear spatially connected to the real environment. A simple overlay is not enough because interaction is required. Analyze the requirement and select the suitable technology.

- A. OCR processing
- B. Mixed reality
- C. Edge filtering
- D. File grouping

Correct Answer: B

Reason:

Mixed Reality supports interaction between real and virtual objects in a shared spatial environment.